

Imperfect Maintenance and Gamma Process

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Problem statement

- Deteriorating system
- Periodic inspections
- Condition-based maintenance
- AGAN corrective maintenance
- Imperfect preventive maintenance

Deterioration model

The deterioration process is a Gamma process, $\{X_t, t \geq 0\}$, with shape function $a(t) = \alpha t^\beta$ with $\alpha > 0$ and $\beta > 0$ and scale parameter $b > 0$. The Gamma process $\{X_t, t \geq 0\}$ is a continuous-time stochastic process such that:

- $X_0 = 0$ with probability 1
- $X_t - X_s \sim G(a(t) - a(s), b)$ for all $t > s \geq 0$, with density function $\forall x \in \mathbb{R}^+$:

$$f(x) = \frac{b^{a(t)-a(s)}}{\Gamma(a(t) - a(s))} x^{a(t)-a(s)-1} e^{-bx}$$

- $\{X_t, t \geq 0\}$ has independent non overlapping increments

Maintenance policy

- $\{Y_t, t \geq 0\}$ the stochastic process of the maintained system.
- Inspection times $(t_j)_{j \in \mathbb{N}}$,
- Periodic inspections $t_j = j\tau, j \in \mathbb{N}$.

Maintenance decision rule

- if $Y_{t_j^-} \geq M$ then a AGAN CM $Y_{t_j^+} = 0$,
- if $L \leq Y_{t_j^-} < M$ imperfect PM, $Y_{t_j^+} = (1 - \rho)Y_{t_j^-}$,
- if $Y_{t_j^-} < L$ no action is performed,

where L and M are fixed thresholds ($0 < L < M$).

By induction, we can conclude that after i inspections without replacement, we have $\forall t \in [i\tau, (i+1)\tau)$

$$Y_t = Y_{i\tau} + (X_t^{(i+1)} - X_{i\tau}^{(i+1)})$$

and if we denote $i = \lfloor t/\tau \rfloor$, then we get

$$Y_t = \sum_{j=1}^i (1-\rho)^{1-U_j} (X_{j\tau}^{(j)} - X_{(j-1)\tau}^{(j)}) + X_t^{(i+1)} - X_{i\tau}^{(i+1)}$$

Finally, we obtain

$$\begin{aligned} Y_{(i+1)\tau} &= Y_{i\tau} + (1-\rho)^{U_{i+1}} (X_{(i+1)\tau}^{(i+1)} - X_{i\tau}^{(i+1)}) \\ &= \sum_{j=1}^{i+1} (1-\rho)^{U_j} (X_{j\tau}^{(j)} - X_{(j-1)\tau}^{(j)}) \end{aligned} \tag{1}$$

- Maintenance policy evaluation : cost criteria
- Maintenance parameter optimisation
- Practical situation: unknown deterioration parameters

Maintenance Cost evaluation

Let C_I , C_P and C_C denote respectively the inspection cost, PM cost and CM cost. The maintenance cost at time t is given as follows:

$$C(t) = C_I \left[\frac{t}{\tau} \right] + C_P \sum_{i=1}^{\lfloor t/\tau \rfloor} \mathbb{1}_{L \leq Y_{i\tau} < M} + C_C \sum_{i=1}^{\lfloor t/\tau \rfloor} \mathbb{1}_{Y_{i\tau} \geq M}$$

As the corrective maintenances are perfect (replacement), by the renewal theory we can compute the long run average maintenance cost as follows:

$$\lim_{t \rightarrow \infty} \frac{C(t)}{t} = \frac{\mathbb{E}(C(T))}{\mathbb{E}(T)} = \frac{C_C + \mathbb{E}(\lfloor T/\tau \rfloor) C_I + \sum_{i=1}^{T/\tau} \mathbb{P}(L \leq Y_{i\tau} < M) C_P}{\mathbb{E}(T)}$$

where T is the length of a renewal cycle which means the period between two corrective maintenance:

$$T = \inf\{t > 0, Y_t \geq M\}$$

To bypass this problem, the markovian renewal property of the maintained degradation process between two inspection is used to calculate the long run average maintenance cost as follows:

$$\lim_{t \rightarrow \infty} \frac{C(t)}{t} = \frac{\mathbb{E}_{\pi}(C(\tau))}{\mathbb{E}_{\pi}(\tau)} = \frac{C_I + C_p \mathbb{P}_{\pi}(L \leq Y_{\tau-} < M) + C_C \mathbb{P}_{\pi}(Y_{\tau-} \geq M)}{\tau}$$

where $\mathbb{E}_{\pi}$ and $\mathbb{P}_{\pi}$ are the expectation and the probability under the steady-state distribution π respectively.

$$\mathbb{P}_{\pi}(L \leq Y_{\tau-} < M) = \int_0^M \int_L^M f(y-x) \pi(dx) dy \quad (2)$$

$$\mathbb{P}_{\pi}(Y_{\tau-} \geq M) = \int_0^{+\infty} \int_M^{+\infty} f(y-x) \pi(dx) dy \quad (3)$$

Stationary law of the Maintained System

Let consider the Markov renewal cycle $[t_{i-1}^-, t_i^-]$, x and y respectively the deterioration level of maintained system at the beginning and the end of the cycle, the steady-state distribution π is the solution of :

$$\pi(y) = \int_0^{+\infty} \pi(x) p(y|x) dx \quad (4)$$

where $p(y|x)$ is the transition law from the state x to the state y . This transition law is equal to

$$p(y|x) = f(y-x)\mathbb{1}_{x < L} + f(y - (1-\rho)x)\mathbb{1}_{L \leq x < M} + f(y)\mathbb{1}_{x > M} \quad (5)$$

where f is the density function of gamma distribution :

$$f(x) = \frac{\beta^{\alpha(t_i^\beta - t_{i-1}^\beta)}}{\Gamma(\alpha(t_i^\beta - t_{i-1}^\beta))} x^{\alpha(t_i^\beta - t_{i-1}^\beta) - 1} e^{-bx} \mathbb{1}_{\mathbb{R}^+}(x)$$

Steady-state distribution of the Maintained System

Thus, we have

$$\begin{aligned}\pi(y) = & \int_0^{\min(L,y)} \pi(x)f(y-x) dx + \int_{\min(\frac{y}{1-\rho},L)}^{\min(\frac{y}{1-\rho},M)} \pi(x)f(y-(1-\rho)x) dx \\ & + f(y) \int_M^{+\infty} \pi(x) dx\end{aligned}$$

We can solve this last equation by fixed-point iteration algorithm by considering that $\pi(x)$ is the solution of $\pi(y) = g(\pi(y))$, where g is a continuous function. Thus, for all y , we approximate π at iteration k by

$$\begin{aligned}w_k(y) = & \int_0^{\min(L,y)} w_{k-1}(x)f(y-x) dx + \int_{\min(\frac{y}{1-\rho},L)}^{\min(\frac{y}{1-\rho},M)} w_{k-1}(x)f(y-(1-\rho)x) dx \\ & + f(y) \int_M^{+\infty} w_{k-1}(x) dx\end{aligned}$$

Stationary Law

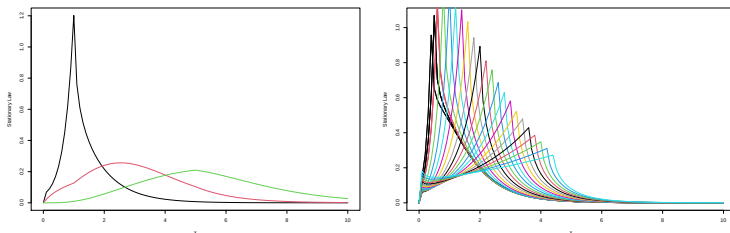


Figure: The steady-state distribution in function of the inspection interval τ (left) and the preventive threshold L (right): $M = 5$, $\rho = 0.8$, $\alpha = 1$, $\beta = 1$, $b = 1$.

Sensitivity analysis

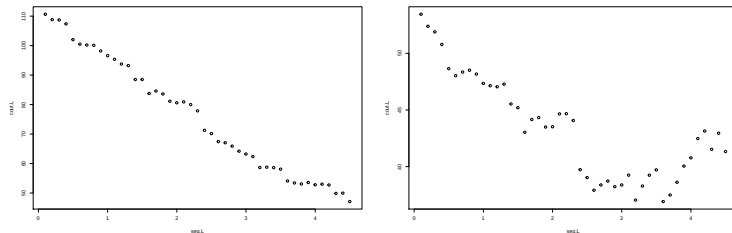


Figure: Maintenance cost in function of the preventive threshold L , with $C_p = \frac{C_c}{2}$ (left) and $C_p = \frac{C_c}{5}$ (right) and $\tau = 0.8$: $M = 5$, $\rho = 0.8$, $\alpha = 1$, $\beta = 1$, $b = 1$.

Sensitivity analysis

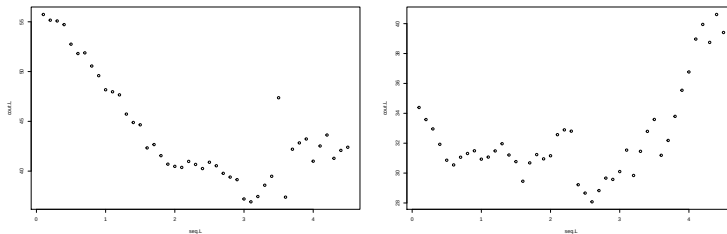


Figure: Maintenance cost in function of the preventive threshold L , with $\tau = 0.5$ (left) and $\tau = 0.8$ (right): $M = 5$, $\alpha = 1$, $\beta = 1$, $b = 1$.

Estimation

Two cases are considered.

- The deterioration level before and after maintenance is known and $\rho \sim \beta(r, s)$.

Estimation by the maximisation of the likelihood of the gamma distributed increments $\{y_{t_j^-} - y_{t_{j-1}^+}, j = 0, 1, \dots\}$.

The maintenance efficiency is estimated by moment method on

$$\frac{y_{t_j^-} - y_{t_{j-1}^+}}{y_{t_j^-}}, \quad j = 1, 2, \dots$$

- The deterioration level is observed only before maintenance. Estimation by the maximisation of the likelihood of the gamma distributed increments $\{y_{t_j^-} - (1 - \rho)y_{t_{j-1}^+}, j = 0, 1, \dots\}$.

The maintenance efficiency is estimated by moment method on

$$\frac{y_{t_j^-} - (1 - \rho)y_{t_{j-1}^+}}{y_{t_j^-}}, \quad j = 1, 2, \dots$$

Thank you for your attention